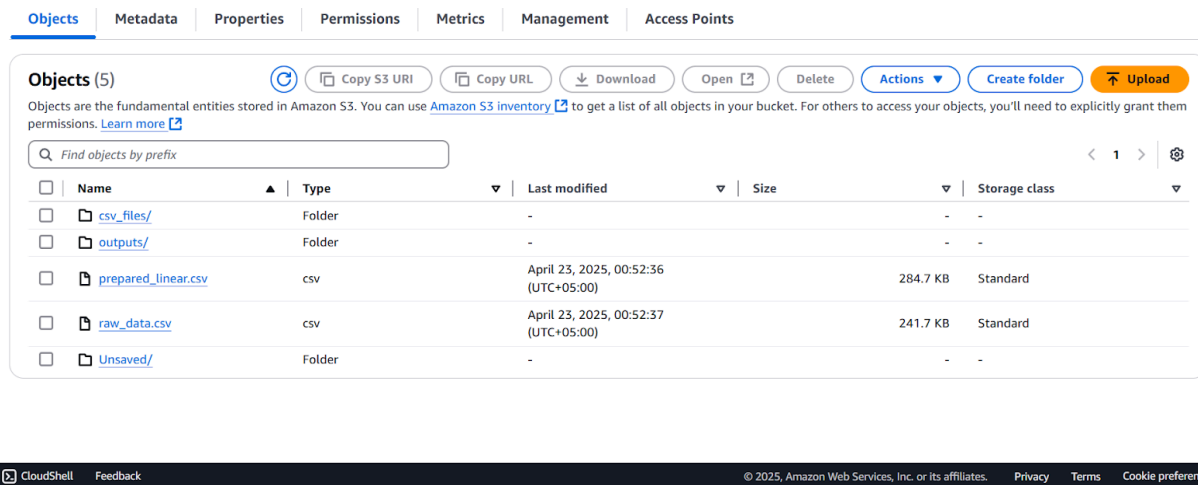
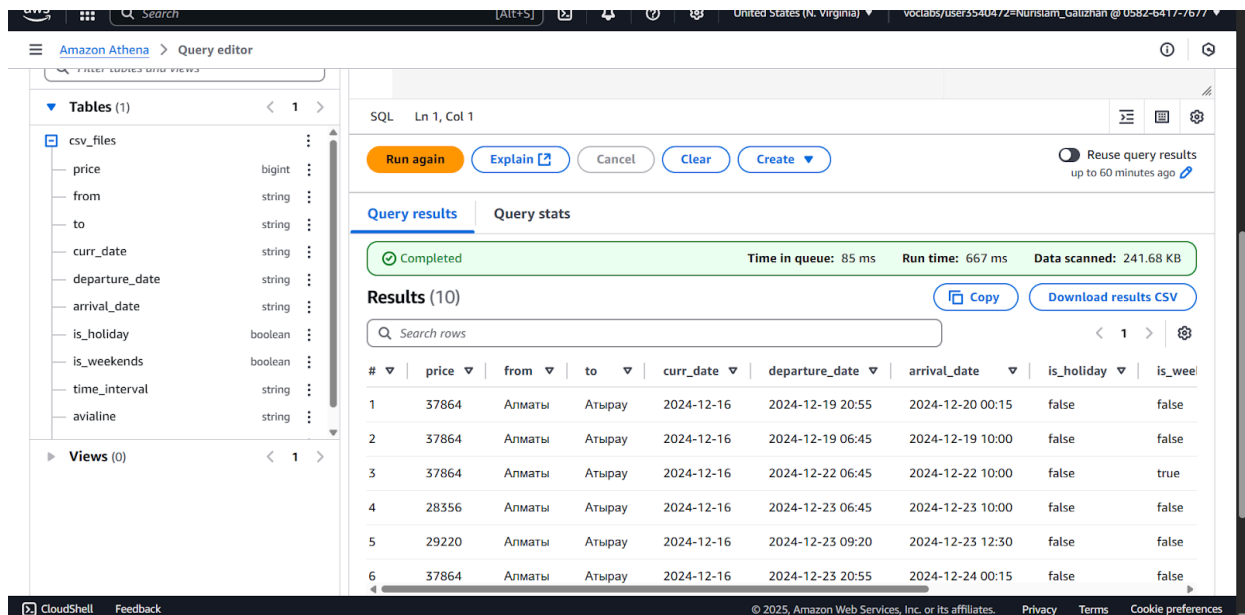


Firstly, we created AWS S3



And after that we created and ran Crawler from AWS Glue. And crawler created table. Which we can check with Athena.



After that we should create ETL Jobs (Add script and notebook) For S3 connection and data preprocessing.

aws Search [Alt+S] United States (N. Virginia) voclabs/user3540472=Nurislam_Galizhan @ 0582-6417-7677

AWS Glue > Jobs

AWS Glue

- Getting started
- ETL jobs**
 - Visual ETL
 - Notebooks
 - Job run monitoring
- Data Catalog tables
- Data connections
- Workflows (orchestration)
- Zero-ETL integrations [New](#)

Data Catalog

- Databases
 - Tables
- Stream schema registries
- Schemas
- Connections
- Crawlers
 - Classifiers
- Catalog settings

Data Integration and ETL

- Zero-ETL integrations [New](#)
- ETL jobs**
 - Visual ETL

AWS Glue Studio Info

Create job [Info](#)

Author in a visual interface focused on data flow. [Visual ETL](#)

Author using an interactive code notebook. [Notebook](#)

Author code with a script editor. [Script editor](#)

[Create example job](#)

Example jobs [Info](#)

Your jobs (1) [Info](#)

Filter jobs by property

☐	Job name	Type	Created by	Last modified	AWS Glue version
☐	test	Glue ETL	Script	23.04.2025, 02:10:34	5.0

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This is the script

test Last modified on 23.04.2025, 02:10:34 [Actions](#) [Save](#) [Run](#)

[Script](#) [Job details](#) [Runs](#) [Data quality](#) [Schedules](#) [Version Control](#)

Script

```
1 import sys
2 from pyspark.context import SparkContext
3 from awsglue.context import GlueContext
4 from awsglue.utils import getResolvedOptions
5 from awsglue.dynamicframe import DynamicFrame
6
7 # Инициализация Glue Context
8 sc = SparkContext()
9 glueContext = GlueContext(sc)
10 spark = glueContext.spark_session
11
12 # Параметры
13 args = getResolvedOptions(sys.argv, ['JOB_NAME'])
14
15 # Чтение данных из S3
16 input_path = "s3://220103263-bucket/csv_files/raw_data.csv"
17 input_data = spark.read.option("header", True).csv(input_path)
```

Python Ln 1, Col 1 Errors: 0 Warnings: 0

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Creating Notebook.

Notebook instances

Info

Actions

Create notebook instance

Q

Search notebook instances

<

1

>

	Name	Instance	Creation time	Status	Actions
☐	my-flight-notebook	ml.t3.medium	4/23/2025, 3:16:20 AM	InService	Open Jupyter Open JupyterLab

And the sugar is in there, creating SageMaker AI, and notebook in it. And code<3

File Edit View Run Kernel Git Tabs Settings Help

+

feature_eng.ipynb

model_training.ipynb

+

Open in...

conda_python3

/ ml_aws /

Name	Modified
feature_eng.ipynb	1s ago
model_training.ipynb	1s ago

[1]:

```
import pandas as pd
import numpy as np

from sklearn.tree import DecisionTreeRegressor
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.svm import SVR
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.linear_model import RidgeCV, LassoCV

from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

[2]:

```
tree_data = pd.read_csv('s3://220103263-bucket/csv_files/processed/prepared_tree.csv')
linear_data = pd.read_csv('s3://220103263-bucket/csv_files/processed/prepared_linear.csv')
```

/home/ec2-user/anaconda3/envs/python3/lib/python3.10/site-packages/fs/spec/registry.py:286: UserWarning: Your installed version of s3fs is very old and known to cause severe performance issues, see also https://github.com/dask/dask/issues/10276

To fix, you should specify a lower version bound on s3fs, or update the current installation.

```
warnings.warn(s3_msg)
```

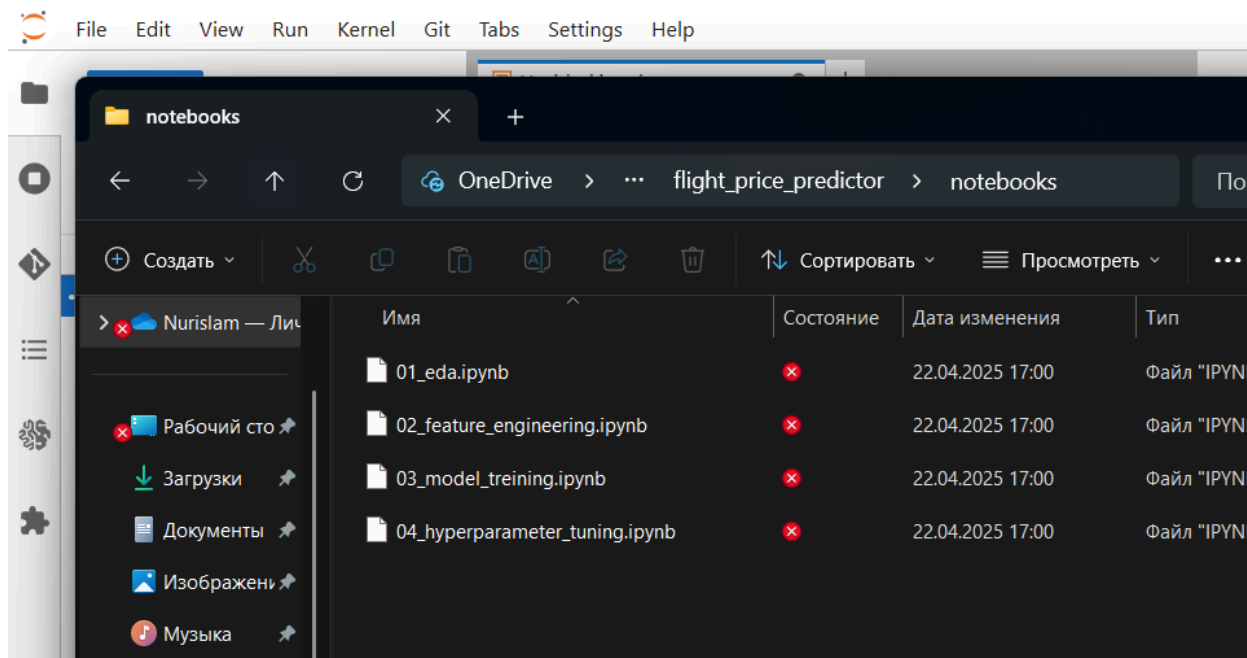
[3]:

```
tree_data
```

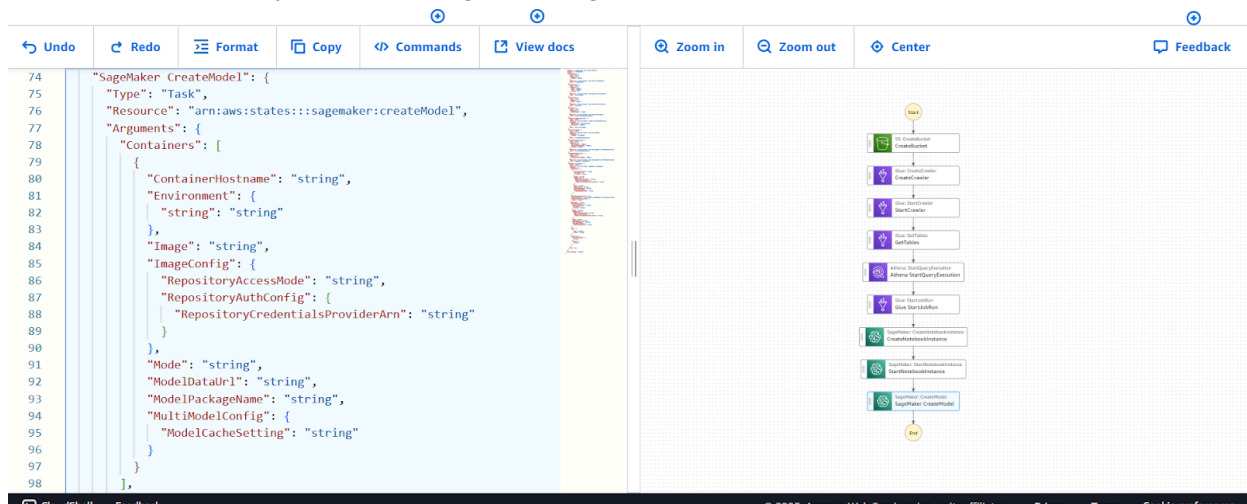
[3]:

	Price	From	To	is_holiday	avialine	duration	days.until flight	part.of day
0	10.131181	0	2	0	1	200	36	1
1	10.069002	0	2	0	1	200	43	1
2	10.069002	0	2	0	1	210	50	2
3	10.069002	0	2	0	1	205	59	1

Save work)



And Also i created my pipeline using QuickSight



My pipeline

Aviasales -> Parser -> CSV_file -> S3 bucket -> Creating Crawler from AWS Glue -> Run Crawler to get Tables -> Check table preview with Athena -> Create ETL Jobs -> Create Notebook in SageMaker AI -> Data Preprocessing and featuring -> Training Model

Architecture Diagram

